# Ward Alkhatib

## Mechatronics Engineering Student, final year · Robotics Trainer

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Mechatronics engineering student at Homs University: five years completed and graduation project presented, graduating in March 2027. Four years teaching robotics to more than 600 children and teens aged 6–18, and coaching competition teams, including Team H2A to silver at the WRO 2023 International Final. I also build and sell ROZE, an Arabic point-of-sale and accounting app for small shops.

## EDUCATION

**Mechatronics, Robotics and Automation Engineering** 2018 – 2027

Faculty of Mechanical and Electrical Engineering, Homs University · five years completed and graduation project presented · expected graduation: March 2027

- **Graduation project, WALL-E Junior** (team of four): an autonomous tracked robot that finds waste, classifies it with YOLOv8 and sorts it with a 5-DOF arm. My part: the computer vision, and the arm and robot design in SolidWorks. In testing it classified 19 of 20 items correctly.
- Control-systems projects modelled and simulated in MATLAB/Simulink. Supervisor of the university's Robotics Club.

**High school diploma (Baccalaureate)**, grade 85% 2018

## EXPERIENCE

**Robotics Trainer & Team Supervisor** · National Center for the Distinguished, Homs Sep 2022 – Sep 2024

- Trained Team H2A and travelled with them to the WRO 2023 International Final in Panama (RoboMission Senior): silver medal. Supervised the center's robotics club.
- Led mechanical design in SolidWorks, and the 3D printing and fabrication of competition robots; taught Arduino programming.

**Robotics Trainer** · Distinction and Creativity Agency, Homs Feb 2022 – Sep 2024

- Taught ages 8–18 with LEGO EV3, SPIKE Prime and Arduino; supervised teams for local competitions and helped organise robotics events.

**Robotics Trainer** · Syra Robot Center, Homs 2020 – Sep 2024

- Taught ages 6–18 with LEGO SPIKE Prime, LEGO EV3, Arduino, Edison and mBot, including children with special needs. Also trained at the Syrian Association for Informatics in As-Suwayda.

**Robotics Club Supervisor** · School for Outstanding Students, Homs 2022

- Trained middle-school students (ages 13–15) in robotics fundamentals.

**Co-founder, technical supervisor and data team leader** · Community Coordination Core, Al-Qurayya Mar 2025 – now

- A youth-led initiative of about 300 university students and graduates serving a town of about 12,000: led a field survey of 2,500 families, built the town database and digital archive, and work with the local committee on services.

**Developer and seller** · ROZE, my own product

- An Arabic point-of-sale and accounting app for small shops, on Windows laptops and Android phones, fully offline. Built in TypeScript with React Native, Expo, SQLite and Electron.

## ACHIEVEMENTS AS A COACH

- **Silver**, WRO 2023 International Final, Panama · Team H2A
- **2 gold**, WRO Syria 2023 (Senior, and Junior for Team Astronauts) · **7 bronze** at WRO Syria and ARC Syria, 2020–2023
- **1st, 2nd and 3rd** in Sumo at ARC Syria 2024, as supervisor · helped organise robotics competitions in Homs governorate

## SKILLS AND MORE

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| **Design and fabrication** | SolidWorks, CAD modelling, design optimisation, 3D printing, mechanical assembly |
| **Electronics and control** | Arduino, sensors, motor control, circuit design · MATLAB, Simulink, PID, MPPT |
| **Software and vision** | Python, YOLOv8 and Ultralytics on Raspberry Pi 5 · TypeScript, React Native, SQLite · MS Office |
| **Educational robotics** | LEGO EV3, LEGO SPIKE Prime, Edison, mBot, Codey Rocky, Scratch |
| **Languages** | Arabic, native · English: reading C1, listening B2, speaking and writing B1 |
| **Volunteering** | Syrian Arab Red Crescent, As-Suwayda, 2016–2018: humanitarian and community service |